Integrala complexă.

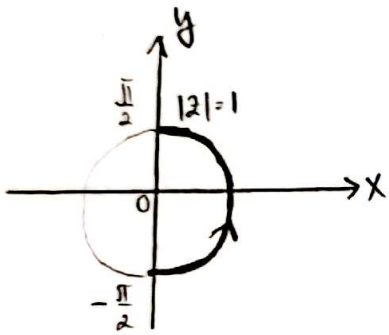
1. Să se calculeze următoarele integrale complexe prin parametrizarea drumurilor indicate.

(a) $\int_\gamma z^2 dz$, unde γ este semicercul $|z|=1$; $\mathrm{Re}(z)\geq 0$ parcurs în sens trigonometric

(b) $\int_\gamma \bar{z}dz$; unde γ este triunghiul cu vârfurile de afix $0, 1, i$, parcurs în sens trigonometric.

Rezolvare

(a)



Semicercul merge de la $-\frac{\pi}{2}$ la $\frac{\pi}{2}$, deci parametrizarea devine:

$\gamma:[-\frac{\pi}{2},\frac{\pi}{2}]\to\mathbb{C}$; $\gamma(t)=z_0+Re^{it}$

$|z|=1 \Rightarrow z_0=0$ și $R=1 \Rightarrow \gamma(t)=e^{it}$

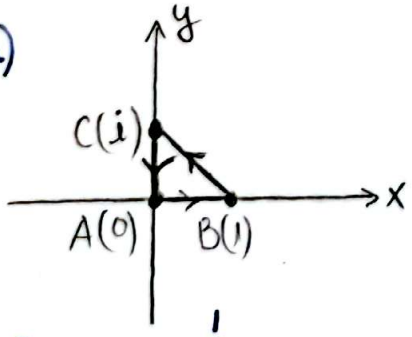
$$\int_\gamma z^2dz=\int_{-\pi/2}^{\pi/2} e^{2it}\cdot i\cdot e^{it}dt = i\int_{-\pi/2}^{\pi/2} e^{3it}dt =$$

$$= i\cdot\frac{e^{3it}}{3i}\Big|_{-\pi/2}^{\pi/2} = \frac{e^{3it}}{3}\Big|_{-\pi/2}^{\pi/2} = \frac{1}{3}\left(e^{\frac{3\pi i}{2}}-e^{-\frac{3\pi i}{2}}\right)=\frac{1}{3}(-i-i)=\frac{-2i}{3}.$$

$e^{\frac{3\pi i}{2}}=\cos\frac{3\pi}{2}+i\sin\frac{3\pi}{2}=0+i\cdot(-1)=-i$

$e^{-\frac{3\pi i}{2}}=\cos\left(-\frac{3\pi}{2}\right)+i\sin\left(-\frac{3\pi}{2}\right)=0+i\cdot 1=i$

(b)



Avem trei segmente orientate, deci putem face trei parametrizări:

$$\int_\gamma \bar{z}dz=\int_{AB}\bar{z}dz+\int_{BC}\bar{z}dz+\int_{CA}\bar{z}dz.$$

AB: $\gamma:[0,1]\to\mathbb{C}$; $\gamma(t)=0+t(1-0)=t$

$$\int_{AB}\bar{z}dz=\int_0^1 t\cdot 1dt=1/2$$

BC: $\gamma:[0,1]\to\mathbb{C}$; $\gamma(t)=1+t(i-1)=1-t+it$

$$\int_{BC}\bar{z}dz=\int_0^1(1-t-it)(i-1)dt=(i-1)\int_0^1(1-t-it)dt=(i-1)\left[\int_0^1 1dt-\int_0^1 tdt+i\int_0^1 tdt\right]=(i-1)\left(1-\frac{1}{2}-\frac{i}{2}\right)=(i-1)(1-i)\cdot\frac{1}{2}=-\frac{1}{2}(i-1)^2=i$$

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CA: $\gamma : [0,1] \to \mathbb{C}$; $\gamma(t) = i + t(-i) = i(1-t)$

$$\int_{CA} \bar{z}\,dz = \int_0^1 i(t-1)\cdot(-i)\,dt = -i\cdot i \int_0^1 (t-1)dt = \frac{t^2}{2} - t\Big|_0^1 = -\frac{1}{2}$$

$$\Rightarrow \int_\gamma \bar{z}\,dz = i$$

2. Să se calculeze următoarele integrale pe contururile închise menționate, presupuse orientate pozitiv:

(a) $\int_{|z|=1} \frac{z^2}{z+3}\,dz$

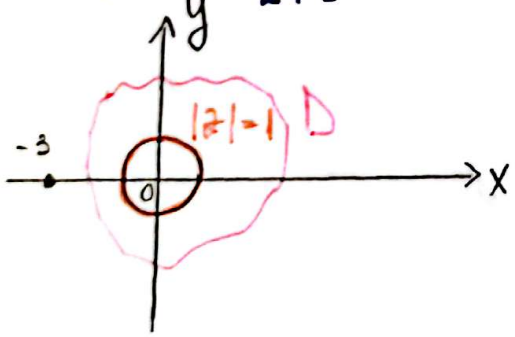
(b) $\int_{|z-1|=1} \frac{1}{z^2+2z+2}\,dz$

(c) $\int_{|z+i|+|z-i|=3} ze^{-z}\,dz$

(d) $\int_{|z-3|=2} \frac{\cos z}{z(z^2+9)}\,dz$.

Rezolvare

(a) $f(z) = \frac{z^2}{z+3}$ olomorfă pe $\mathbb{C}\setminus\{-3\}$



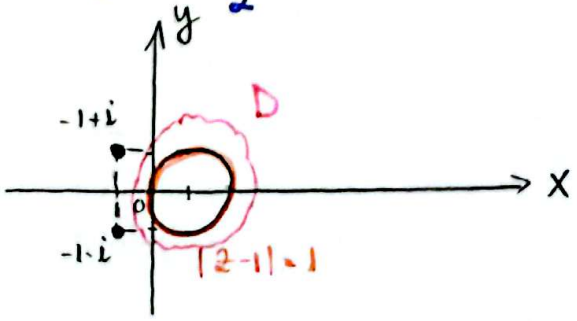
$\gamma: |z|=1$

$\exists D$ domeniu a.î. f olom pe D
$-3 \notin D$
$K \subseteq D$

$\Rightarrow \int_{\partial K} f(z)\,dz = 0$

(b) $f(z) = \frac{1}{z^2+2z+2}$ olomorfă pe $\mathbb{C}\setminus\{-1\pm i\}$

$z^2+2z+2 = 0$
$\Delta = 4-8 = -4 \Rightarrow \sqrt{\Delta} = 2i$
$z_{1,2} = \frac{-2\pm 2i}{2} = -1\pm i$



$\gamma: |z-1|=1$
$\exists D$ domeniu a.î. f olomorfă pe D
$-1\pm i \notin D$
$K \subseteq D$

$\Rightarrow \int_{\partial K} f(z)\,dz = 0$

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(c) $f(z) = e^{-z} \cdot z \cdot \frac{z}{e^z}$ olomorfă pe $\mathbb{C}$

$|z+i| + |z-i| = 3 \Leftrightarrow |x+iy+i| + |x+iy-i| = 3 \Leftrightarrow$

$\Leftrightarrow \sqrt{x^2+(y+1)^2} + \sqrt{x^2+(y-1)^2} = 3 \Leftrightarrow \sqrt{x^2+(y+1)^2} = 3 - \sqrt{x^2+(y-1)^2} \uparrow^2 \Leftrightarrow$

$\Leftrightarrow x^2 + (y+1)^2 = 9 + x^2 + (y-1)^2 - 6\sqrt{x^2+(y-1)^2} \Leftrightarrow$

$\Leftrightarrow y^2 + 2y + 1 = 9 + y^2 - 2y + 1 - 6\sqrt{x^2+(y-1)^2} \Leftrightarrow$

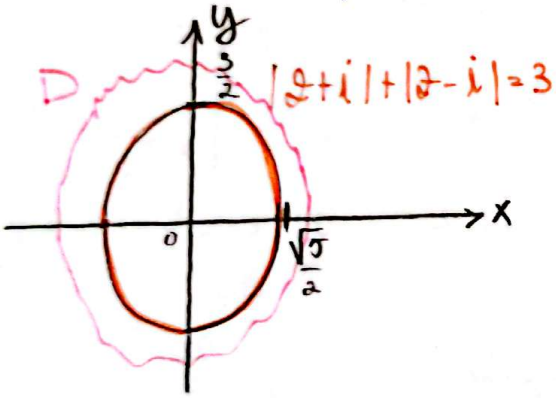
$\Leftrightarrow 6\sqrt{x^2+(y-1)^2} = 9 - 4y \uparrow^2 \Leftrightarrow$

$\Leftrightarrow 36(x^2+(y-1)^2) = 81 + 16y^2 - 72y \Leftrightarrow$

$\Leftrightarrow 36x^2 + 36y^2 - 72y + 36 = 81 + 16y^2 - 72y \Leftrightarrow$

$\Leftrightarrow 36x^2 + 20y^2 = 45 \mid :45 \Leftrightarrow$

$\Leftrightarrow \frac{x^2}{\left(\frac{\sqrt{5}}{2}\right)^2} + \frac{y^2}{\left(\frac{3}{2}\right)^2} = 1$ (elipsă).

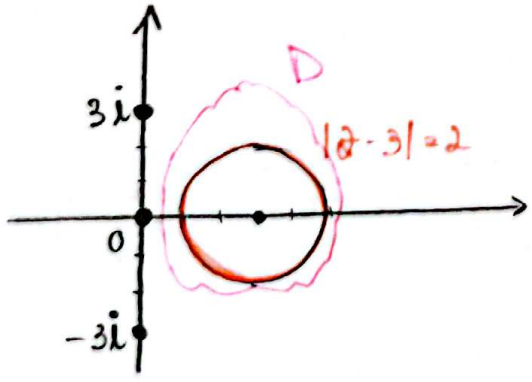


$f: |z+i| + |z-i| = 3$

$\exists D$ domeniu a.s. f olomorfă pe D, $D \supseteq K$ $\Rightarrow$

$\Rightarrow \int_{\partial K} f(z)dz = 0$

(d) $f(z) = \frac{\cos z}{z(z^2+9)}$ olomorfă pe $\mathbb{C} \setminus \{0; \pm 3i\}$



$f: |z-3| = 2$

$\exists D$ domeniu a.s. f olomorfă pe D, $0, \pm 3i \notin D$, $K \subseteq D$ $\Rightarrow$

$\Rightarrow \int_{\partial K} f(z)dz = 0$

3. Să se calculeze următoarele integrale pe contururile închise menționate, presupuse orientate pozitiv:

(a) $\int_{\gamma} \frac{e^{-z}}{22-5i}dz$, unde γ este pătratul cu laturile situate pe dreptele $x = \pm 2$; $y = \pm 2$.

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(b) $\int_{|z|=1} \frac{\cos z}{z(z^2+8)}\,dz$

(c) $\int_{|z|+|z-1|=2} \frac{3z-2}{z^2-z}\,dz$

(d) $\int_{|z|=1} \frac{e^{2z}}{z^4}\,dz$

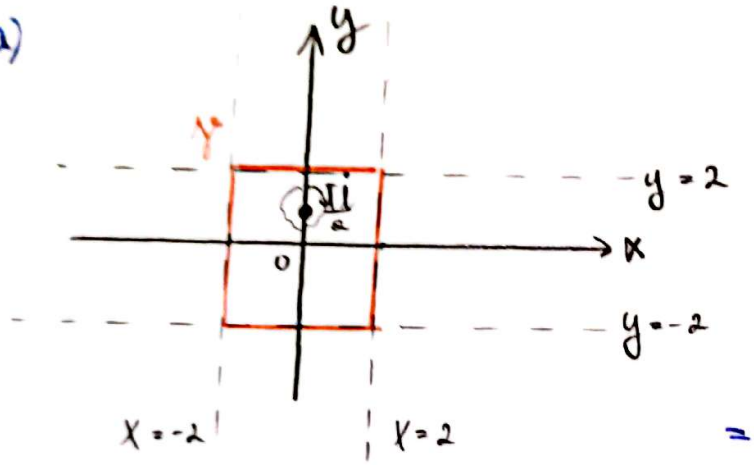
(e) $\int_{|z-i|=2} \frac{1}{(z^2+4)^2}\,dz$

(f) $\int_{|z-i|=2} \frac{2z^2-z+1}{(z-1)^2(z+1)}\,dz$

(g) $\int_{|z|=4} \frac{e^z}{(z^2+\pi)^2}\,dz.$

Rezolvare

(a)



$$\int_\gamma \frac{e^{-z}}{2z-\pi i}\,dz = \int_\gamma \frac{\overbrace{\frac{1}{2}\cdot e^{-z}}^{g(z)}}{z-\frac{\pi i}{2}}\,dz$$

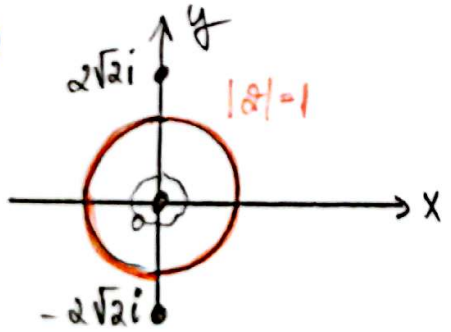
Cauchy $2\pi i\, g\left(\frac{\pi i}{2}\right) =$

$= 2\pi i \cdot \frac{1}{2} e^{-z}\Big|_{z=\frac{\pi i}{2}} =$

$= \pi i\, e^{-\frac{\pi i}{2}} = \pi i\left(\cos\frac{\pi}{2} - i\sin\frac{\pi}{2}\right) =$

$= \pi i(0 - i\cdot 1) = -i^2\pi = \pi.$

(b)



$$\int_{|z|=1} \frac{\cos z}{z(z^2+8)}\,dz = \int_{|z|=1} \frac{\overbrace{\frac{\cos z}{z^2+8}}^{g(z)}}{z}\,dz =$$

Cauchy $2\pi i\, g(0) = 2\pi i \cdot \frac{\cos z}{z^2+8}\Big|_{z=0} =$

$= 2\pi i \cdot \frac{1}{8} = \frac{\pi i}{4}$

(c) $|z|+|z-1| = 2 \Leftrightarrow |x+iy| + |x+iy-1| = 2 \Leftrightarrow$

$\Leftrightarrow \sqrt{x^2+y^2} + \sqrt{(x-1)^2+y^2} = 2 \Leftrightarrow \sqrt{x^2+y^2} = 2 - \sqrt{(x-1)^2+y^2}\ \uparrow^2 \Leftrightarrow$

$\Leftrightarrow x^2+y^2 = 4+(x-1)^2+y^2 - 4\sqrt{(x-1)^2+y^2} \Leftrightarrow$

$\Leftrightarrow x^2 = 4+x^2-2x+1 - 4\sqrt{(x-1)^2+y^2} \Leftrightarrow$

$\Leftrightarrow 4\sqrt{(x-1)^2+y^2} = 5-2x\ \uparrow^2 \Leftrightarrow$

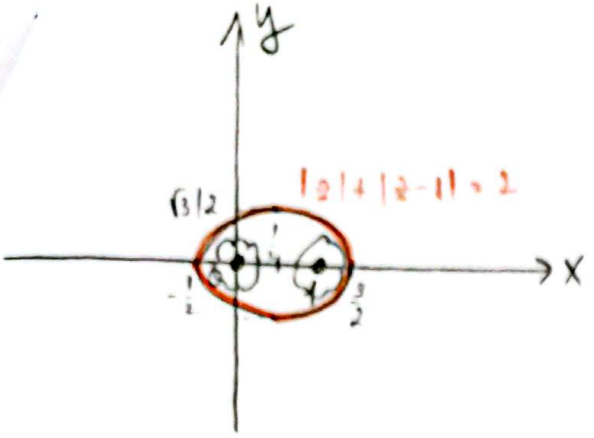
$\Leftrightarrow 16(x-1)^2+16y^2 = 25+4x^2-20x \Leftrightarrow$

$\Leftrightarrow 16x^2-32x+16+16y^2 = 25+4x^2-20x \Leftrightarrow$

$\Leftrightarrow 12x^2-12x+16y^2 = 9 \Leftrightarrow$

$\Leftrightarrow 12\left(x-\frac{1}{2}\right)^2+16y^2 = 12 \Leftrightarrow \frac{\left(x-\frac{1}{2}\right)^2}{1^2} + \frac{y^2}{\left(\frac{\sqrt{3}}{2}\right)^2} = 1$ (elipsă).

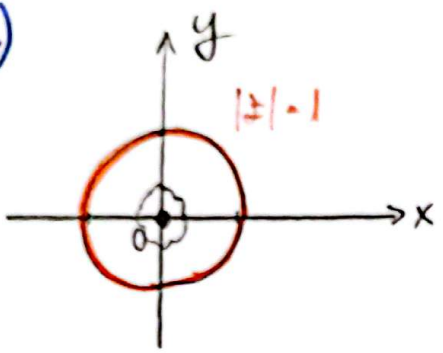
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$$\int_{|z|+|z-1|=2} \frac{3z-2}{z^2-z}\,dz = \int_{\gamma_1} \frac{\frac{3z-2}{z}\cdot g(z)}{z-1}\,dz + \int_{\gamma_2} \frac{\frac{3z-2}{z-1}\cdot g(z)}{z}\,dz \quad \text{Cauchy}$$

$$= 2\pi i\cdot(f(1)+g(0)) = 2\pi i\,(1+2) = 6\pi i.$$

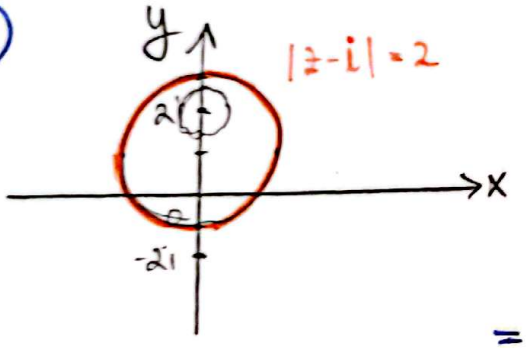
(d)



$$\int_{|z|=1} \frac{e^{z\cdot 2}\cdot g(z)}{z^4}\,dz \quad \text{Cauchy}$$

$$= \frac{2\pi i}{3!}\cdot g'''(0) = \frac{\pi i}{3}\cdot 8\cdot e^{2z}\Big|_{z=0} = \frac{8\pi i}{3}\cdot e^{2\cdot 0} = \frac{8\pi i}{3}.$$

(e)

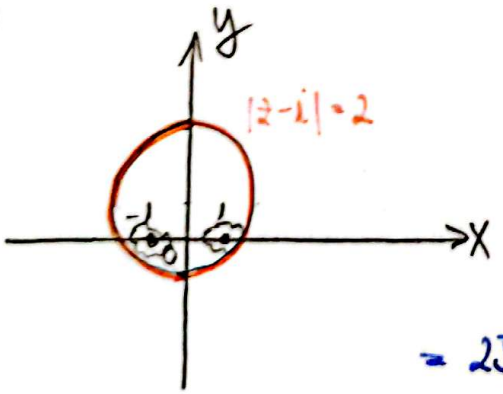


$$\int_{|z-i|=2} \frac{1}{(z^2+4)^2}\,dz = \int_{|z-i|=2} \frac{1}{(z-2i)^2(z+2i)^2}\,dz$$

$$= \int_{|z-i|=2} \frac{\frac{1}{(z+2i)^2}\cdot g(z)}{(z-2i)^2}\,dz = \frac{2\pi i}{1!}\,g'(z)\Big|_{z=2i}$$

$$= 2\pi i\cdot\frac{-2}{(z+2i)^3}\Big|_{z=2i} = -4\pi i\cdot\frac{1}{(4i)^3} = \frac{\pi}{16}$$

(f)



$$\int_{|z-i|=2} \frac{2z^2-z+1}{(z-1)^2(z+1)}\,dz$$

$$= \int_{\gamma_1} \frac{\frac{2z^2-z+1}{(z-1)^2}\cdot g(z)}{z+1}\,dz + \int_{\gamma_2} \frac{\frac{2z^2-z+1}{z+1}\cdot g(z)}{(z-1)^2}\,dz$$

Cauchy $2\pi i\,f(-1) + \frac{2\pi i}{1!}\cdot g'(1) =$

$$= 2\pi i\left(1 + \frac{(4z-1)(z+1) - 2z^2+z-1}{(z+1)^2}\right)\Big|_{z=1}$$

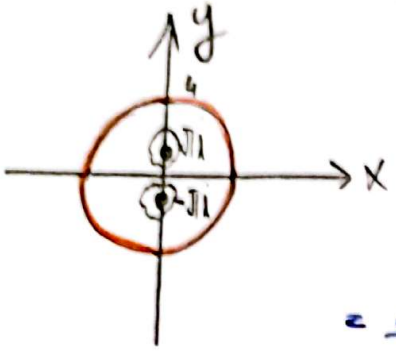
$$= 2\pi i\left(1 + \frac{4z^2+4z-1-2z^2+z-1}{(z+1)^2}\right)\Big|_{z=1}$$

$$= 2\pi i\left(1 + \frac{2z^2+4z-2}{(z+1)^2}\right)\Big|_{z=1}$$

$$= 2\pi i\left(1+1\right) = 4\pi i$$

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(g)



$$\int_{|z|=4} \frac{e^z}{(z^2+\pi^2)^2}dz = \int_{|z|=4} \frac{e^z}{(z+\pi i)^2(z-\pi i)^2}dz =$$

$$= \int_{\gamma_1} \frac{\overbrace{e^z}^{f(z)}}{\underbrace{(z+\pi i)^2}_{}\,'} dz + \int_{\gamma_2} \frac{\overbrace{e^z}^{g(z)}}{\underbrace{(z-\pi i)^2}_{}\,'} dz \quad \text{Cauchy}$$

$$= \frac{2\pi i}{11}\left(f'(\pi i) + g'(-\pi i)\right) =$$

$$= 2\pi i \cdot \frac{e^z(z+\pi i)^2 - e^z\cancel{(z+\pi i)}\cdot 2}{(z+\pi i)^{4\,3}}\Bigg|_{z=\pi i} + 2\pi i \cdot \frac{e^z(z-\pi i)^2 - e^z\cdot 2\cancel{(z-\pi i)}}{(z-\pi i)^{4\,3}}\Bigg|_{z=-\pi i}$$

$$= 2\pi i \cdot \frac{e^z(z+\pi i-2)}{(z+\pi i)^3}\Bigg|_{z=\pi i} + 2\pi i \cdot \frac{e^z(z-\pi i-2)}{(z-\pi i)^3}\Bigg|_{z=-\pi i} =$$

$$= 2\pi i \cdot \frac{e^{\pi i}\cdot 2(\pi i-1)}{(2\pi i)^3} + 2\pi i \cdot \frac{e^{-\pi i}\cdot(-2)(\pi i+1)}{(-2\pi i)^3} =$$

$$= \frac{-1\cdot 2(\pi i-1)}{(2\pi i)^2} - \frac{-1\cdot(-2)(\pi i+1)}{(2\pi i)^2} = -\frac{2\pi i - \cancel{2} + 2\pi i + \cancel{2}}{(2\pi i)^2} =$$

$$= \frac{-4\pi i}{-4\pi^2} = \frac{i}{\pi}$$

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